



FireCast-fusion: Physics-Guided fusion of UAV RGB–thermal imagery and environmental data for near-term wildfire spread prediction

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ABSTRACT

Accurately forecasting near-term wildfire spread from aerial imagery is critical for emergency response. However, it remains challenging because fire growth is driven by complex factors like wind, terrain, and fuel variability. Most prior work focused on per-frame fire detection or mapping rather than true spread prediction and often ignored environmental dynamics. As a result, purely data-driven models have struggled to generalize and can produce physically inconsistent fire spread forecasts under changing wind or terrain conditions. To address these issues, we propose FireCast-Fusion, a deep learning framework that fuses multimodal UAV imagery with environmental data for short-horizon wildfire forecasting. The model integrates a temporal transformer encoder (TTE) to capture the evolving fire-front dynamics from sequential UAV frames, and a physics-guided diffusion (PGDL) module that incorporates wind direction and slope information to enforce realistic spread behavior. By combining visual cues with physical constraints, FireCast-Fusion produces both probabilistic fire-front maps and per-pixel arrival-time estimates for the advancing wildfire. The system is trained with a multi-objective loss that jointly optimizes fire segmentation accuracy, arrival-time prediction, and physical consistency, ensuring the model learns to balance predictive performance with adherence to real-world spread dynamics. Experimental results show that our approach outperforms conventional 3D CNN and LSTM-based models, improving fire-front intersection-over-union and arrival-time error by up to 18%. The hybrid transformer–diffusion design also yields more stable predictions under varying wind conditions and rugged topography, producing interpretable spread trajectories that align closely with observed fire behavior.

1. Introduction

Wildfires are becoming more frequent and severe, driven by rising temperatures and extended dry seasons (Aibin et al., 2024). Accurate short-term spread forecasting is vital for protecting human life and infrastructure, yet current systems remain limited in resolution, interpretability, and real-time adaptability (Chen et al., 2022). Traditional remote-sensing methods based on satellite imagery provide wide coverage but lack the fine spatial and temporal granularity needed to capture sub-meter fire fronts or fast-changing local conditions. Unmanned Aerial Vehicles (UAVs) have emerged as a more responsive alternative, capable of recording high-resolution RGB and thermal imagery that reflects dynamic fire behavior, wind interactions, and terrain influence in real time (Zhang et al., 2025; Li et al., 2024; Hu et al., 2025).

Despite these advances, a fundamental limitation persists in current

wildfire modeling approaches. Existing systems are typically designed either for visual fire detection or for physics-based simulation, but rarely for unified spatiotemporal prediction. This separation leads to a critical gap: models that rely solely on visual data fail to capture the underlying physical drivers of fire propagation, while physics-based methods lack the flexibility and scalability required for real-time deployment on UAV platforms. As wildfire spread is inherently governed by complex interactions between environmental conditions and observed fire dynamics, an effective forecasting system must jointly model both aspects in a coherent framework.

Existing deep learning frameworks for wildfire analysis have primarily focused on fire detection or static burned-area mapping. Convolutional neural networks (CNNs) and hybrid ViT–CNN models have improved accuracy under smoke or low visibility (Pesonen et al., 2025; Kim et al., 2025; Arlović et al., 2025), but they remain limited by their

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reliance on visual appearance. These models often ignore environmental variables such as slope, wind, and vegetation density, leading to overfitting, poor generalization across terrains, and inconsistent fire-front boundaries when smoke or lighting conditions vary. Moreover, most available UAV datasets lack temporal annotations like per-pixel arrival time or fuel-type maps, making it difficult for models to learn how fires evolve over time. As a result, many existing systems provide visually accurate detections (Wang et al., 2025; Yang et al., 2025; Tian et al., 2025) but fail to predict future fire progression or capture physically realistic spread dynamics.

Physics-based simulators such as Cell2Fire and cellular automata methods offer physically interpretable spread predictions, but they require detailed environmental inputs and are computationally expensive for real-time UAV deployment (Marques et al., 2024; Zhuang et al., 2025; Qiao et al., 2024). Meanwhile, purely data-driven deep networks can learn complex correlations but often violate physical constraints, predicting fire propagation that ignores wind direction or slope influence. The lack of integration between learned visual cues and physical dynamics remains a major gap in current research. Efforts to incorporate physical knowledge through Physics-Informed Neural Networks (PINNs) have improved generalization but struggle to process the high-dimensional, multimodal UAV imagery needed for operational forecasting.

Recent advances in geostationary satellite-based wildfire detection using Himawari-8/9 have demonstrated near real-time monitoring capabilities at approximately 10-min intervals through multi-feature fusion and transformer-based temporal modeling (Ma et al., 2025; Kang et al., 2022; Zhang et al., 2023). These approaches effectively capture large-scale ignition patterns and early fire signals by integrating spectral, spatial, and temporal information. However, they operate at coarse spatial resolution and primarily focus on detection rather than fine-grained wildfire spread prediction. In contrast, the proposed FireCast-Fusion framework targets high-resolution UAV imagery and models short-horizon fire-front propagation using physics-guided diffusion constraints.

To address these challenges, this study introduces FireCast-Fusion, a hybrid physics-guided transformer (Zhang and Wang, 2025; Han et al., 2025; Liu et al., 2025) framework that unites data-driven visual learning with physically consistent modeling. The proposed system integrates UAV RGB–thermal sequences with environmental layers such as wind vectors, slope, elevation, and vegetation indices to create a comprehensive spatiotemporal representation of fire behavior. A Temporal Transformer Encoder (Liu et al., 2025; Yu et al., 2023, 2025; Hassanin et al., 2025) captures long-range spatial and temporal dependencies of fire motion, while a Physics-Guided Diffusion Layer enforces physical spread constraints through an advection–diffusion process that aligns fire propagation with local wind and terrain gradients. Together, these components produce accurate fire probability maps and arrival-time estimations, bridging the gap between visual perception and physical reasoning. While satellite-based systems provide early large-scale detection, they lack the spatial resolution and temporal modeling required for fine-grained wildfire spread prediction from UAV observations.

1.1. Research significance

The main contributions of FireCast-Fusion are:

1. A multimodal framework integrating UAV RGB–thermal imagery with environmental factors such as wind, terrain, and vegetation for wildfire spread prediction.
2. A hybrid architecture combining a temporal transformer with a physics-guided diffusion layer to capture spatiotemporal dynamics and enforce physically consistent propagation.
3. A dual-output design for simultaneous estimation of fire probability and per-pixel arrival time.

4. A unified multi-objective training strategy balancing segmentation accuracy, temporal consistency, and physical constraints.
5. Improved performance over baseline models across multiple UAV datasets, demonstrating robust generalization and stable prediction under varying conditions.

1.2. Paper organization

This paper is organized into seven sections. Section 1 introduces the challenges of short-term wildfire spread forecasting, emphasizing the gaps in current image-only and simulation-based models. Section 2 reviews related work across four key areas: UAV wildfire datasets, fire detection and segmentation with deep learning, physics-guided fire modeling, and multimodal prediction frameworks. Section 3 describes the proposed approach, detailing its temporal transformer encoder, physics-guided diffusion layer, and multi-objective training scheme. Section 4 outlines the experimental setup, including dataset descriptions, input preprocessing, model configuration, and evaluation metrics. Section 5 presents quantitative and qualitative results, comparing FireCast-Fusion against state-of-the-art baselines on spread accuracy and segmentation fidelity. Section 6 discusses the model's strengths and limitations, deployment considerations, and robustness to different terrains. Section 7 concludes with a summary of contributions.

2. Related works

2.1. UAV and remote sensing datasets for wildfire analysis

Recent advances in UAV sensing have provided high-resolution, multi-modal imagery for localized fire detection. Zhang et al. (2025) released a multi-scenario UAV dataset with time-synchronized RGB and thermal images (RGBT-3M) to capture both visible flames and heat signatures. By fusing these complementary modalities, their YOLO-based model achieved over 92% precision, demonstrating improved early fire detection under smoke and lighting variability. Hu et al. (Li et al., 2024) introduced UAV-StrawFire, a visible–IR video dataset from controlled straw burns, annotated for flame detection, segmentation, and tracking. Using this data, they developed lightweight models (e.g. FF-YOLOv5n and BiSeNetV2) that reliably delineated fire regions (mIoU >0.88) and tracked fire spread at 50 FPS, enabling real-time on-board monitoring. Pesonen et al. (2025) addressed the data scarcity by releasing the Boreal Fire UAV dataset with hundreds of geo-annotated images and video clips from prescribed burns in Finnish forests. This dataset includes aligned RGB images with smoke/flame bounding boxes, segmentation masks, and labeled video frames, providing a valuable open benchmark for training both detection and segmentation models. Similarly, Mowla et al. (2024) presented the UAVs-FFDB database comprising 1653 high-resolution UAV images (augmented to over 15,000 images) of early-stage fires. The UAVs-FFDB dataset greatly expands the diversity of fire scenes and has been meticulously annotated for flame boundaries, facilitating research on prompt fire recognition. These efforts have substantially improved visual coverage (day/night, smoky conditions) and data volume for model training. However, most UAV fire datasets still provide only frame-level labels (classification or masks) without per-pixel ignition times or temporal sequences. This gap in spatiotemporal annotations limits their direct use for spread forecasting and motivates data fusion with physics models or synthetic sequences in subsequent work.

2.2. Deep learning methods for fire detection and mapping

Early deep-learning approaches to wildland fire focused on frame-wise fire detection or segmentation from aerial images. Traditional CNN-based models (e.g. Fire-Net) achieved high accuracy and real-time speed on single-frame detection tasks (Hu et al., 2025). For instance, Li et al. (Hu et al., 2025) proposed FireNet, a specialized lightweight CNN

that integrates a tiny Transformer module (RepViT) for global context and a decoupled head for flame/smoke localization. FireNet runs in ~27 ms per frame and improved small-fire detection precision to ~82.6%, outperforming YOLOv8 while remaining feasible for drone deployment. Other efficient CNN variants, such as a residual capsule-attention network by Othman and Ali (Vogiatzoglou et al., 2025), have further enhanced robustness to challenging conditions (e.g. dense smoke or complex backgrounds) in UAV imagery. However, these 2D models process each frame independently and lack temporal memory. They often struggle with fragmented fire masks or missed early ignitions because sequential context (growth trends, motion cues) is not utilized. To address this, researchers have explored attention-based architecture.

Advances in AI-driven wildfire monitoring using remote sensing have further demonstrated the potential of deep learning for large-scale fire detection and risk assessment. For example, recent studies in Scientific Reports have applied convolutional and hybrid deep-learning models to satellite-based fire monitoring and environmental risk estimation under diverse geographic conditions (Illarionova et al., 2025; Shadrin et al., 2024). These works primarily focus on regional-scale fire detection and classification using multispectral satellite imagery. In contrast, FireCast-Fusion addresses short-horizon wildfire spread prediction using high-resolution UAV RGB-thermal sequences and integrates physics-guided diffusion constraints to model directional fire-front propagation at fine spatial scales.

Zhou et al. (2025) compared standard CNNs, hybrid CNN-RNNs, and pure Transformers for wildfire image analysis, and found that multi-head self-attention consistently improved multi-scale feature extraction and reduced false alarms in smoke-like backgrounds (e.g. cloud, fog) by learning global context. Qiao et al. (2024) went further, applying a Vision Transformer to infer ignition points from final wildfire perimeters. Their model effectively captured long-range spatial relationships in burn scars to predict where the fire started. These attention-based methods outperform frame-bound CNNs on spatial coherence and early-warning capability. Nonetheless, vision models alone remain appearance driven. They can be fooled by visual ambiguities (sun glare, smoke shadows) and may produce inconsistent spreading maps under changing environmental conditions. To increase reliability, some works combined learned inference with rule-based constraints. For example, Dubey et al. (Li et al., 2022) developed a decision-support system that uses a Transformer to forecast fire spread over time while a fuzzy rule-base imposes expert constraints (e.g. if wind shifts, slow the lateral spread). This hybrid approach improved the realism of large-scale satellite fire predictions, but it was tested at coarse (kilometer-level) resolution and on relatively static scenarios. In summary, modern deep models, especially Transformers have greatly advanced fire detection and mapping from imagery, reducing false positives and improving small-fire and smoke recognition. Yet purely data-driven models still lack an inherent understanding of fire physics or temporal progression, motivating integration of physical knowledge as discussed next.

Recent advances in geostationary satellite-based wildfire detection have demonstrated strong capabilities for early-stage fire monitoring at high temporal frequency. Systems based on Himawari-8/9 satellites achieve near real-time detection at 10-min intervals by integrating spectral, spatial, and temporal features (Ma et al., 2025; Kang et al., 2022; Zhang et al., 2023). Specifically, multi-feature fusion approaches combine thermal anomalies and contextual spatial patterns for robust detection (Ma et al., 2025), while recursive transformer-based frameworks model temporal dependencies and multi-source information for early ignition detection (Kang et al., 2022). In addition, methods that jointly exploit spatial, temporal, and spectral information improve sensitivity to weak fire signals during early-stage events (Zhang et al., 2023).

However, these satellite-based approaches primarily operate at coarse spatial resolution and are optimized for fire detection rather than fine-grained spread prediction. In contrast, the proposed FireCast-Fusion framework focuses on high-resolution UAV imagery and short-horizon

wildfire propagation modeling. By integrating multimodal visual inputs with environmental drivers and physics-guided diffusion constraints, our approach complements satellite-scale detection systems by enabling localized, physically consistent fire spread forecasting.

2.3. Physics-guided and surrogate modeling approaches

An alternative line of research has embedded physical wildfire dynamics into learning frameworks (Purcell et al., 2025). Rather than relying solely on visual cues, these methods incorporate fire-spread physics (combustion, diffusion, advection) to improve realism. Kim et al. (2025) recently benchmarked a cellular automata simulator called Cell2Fire. It is a simplified fire spread model against standard operational models in the US, Canada, and Chile. They showed Cell2Fire can efficiently reproduce fire front propagation and burn patterns under various fuel and terrain conditions. By applying multi-objective optimization to Cell2Fire's parameters, Kim et al. improved simulation accuracy on a real Alberta wildfire (raising F1score from 0.74 to 0.83) while maintaining faster-than-real-time runtimes. This demonstrates the promise of data-driven tuning of physics simulators. Moving toward learning-based surrogates, Vogiatzoglou et al. (Zhang and Wang, 2025) used physics-informed neural networks (PINNs) to learn key fire spread parameters directly from observational data. Their PINN model produced interpretable, physically consistent spread predictions and could estimate unknown rate-of-spread parameters from partial burn maps. However, it required continuous spatial fields (e.g. temperature, fuel moisture) for training, which UAV imagery alone rarely provides. Yarmohammadian et al. (2025) performed a comprehensive review of physics-informed surrogate models in wildfire and fire safety. They concluded that embedding even simplified physical priors (energy conservation, diffusion equations) into neural networks markedly enhances realism and helps prevent overfitting to scene-specific textures. In practice, integrating physics can constrain a model to feasible fire behaviors, addressing the “physics-free” nature of standard deep learning. For example, learned cellular automata approaches have yielded strong results: Zhuang et al. (2025) trained a data-driven CA model on multi-temporal satellite/UAV images and achieved over 90% accuracy in predicting burned areas. Similarly, Tong et al. (2024) showed that neural networks can learn to predict arrival time maps from simulated fires, effectively parameterizing spread dynamics and exceeding 90% accuracy in burned-area estimation.

Building on these ideas, Zhou et al. (2025) extended physical fusion to modern architecture by training a Transformer to adapt its spread kernel based on wind and slope inputs (analogous to learning an internal ROS formula). Their transformer-driven model automatically adjusted fire growth directionality with changing wind/slope, outperforming fixed-rule CA simulators in complex terrain. Furthermore, researchers have emphasized coupling AI with full physics simulators. Caron et al. (Tenny et al., 2025) and Tutland et al. (2025) argue for hybrid systems where deep models leverage 3D fuel data and heat transfer simulations to ensure consistency with real fire behavior. For instance, incorporating LiDAR-derived canopy fuel distributions or using a CFD fire model to guide a neural network can significantly improve the interpretability of predictions in heterogeneous forests. In summary, the infusion of physical principles whether via simplified CA rules, PINN loss constraints, or simulator-in-the-loop training has proven critical for modeling wildfire spread. These methods produce more realistic and temporally consistent fire progression forecasts than vision-only models, though they often depend on simulated data or external environmental inputs that are not always available in UAV operations.

2.4. Synthetic data, uncertainty, and generative modeling

Robustness across diverse wildfire scenarios remains a major challenge for purely data-trained models. Researchers have turned to synthetic data augmentation and generative AI to address limited real data

and occlusion issues. Wang et al. (2025) introduced a large Multi-Faceted Synthetic Wildfire Dataset (M4SFWD) constructed with a game engine. The M4SFWD dataset spans multiple forest types, terrains, lighting conditions, and weather scenarios, providing thousands of labeled fire images under controlled variations. Models pre-trained on this synthetic data and then fine-tuned on real wildfire images showed improved generalization. In effect, the synthetic pre-training acted as a regularizer against smoke, lighting, and scale changes. Arlović et al. (2025) quantified the impact of such synthetic augmentation on fire segmentation performance. By blending their SYN-FIRE indoor fire images with real datasets, they observed consistent Dice score improvements of 2–16% across multiple test sets, with even small injections of synthetic data boosting model generality. Notably, replacing real images with synthetic ones could still improve segmentation in some cases, though synthetic training regimes risk missing real-world nuances. The consensus is that moderate synthetic augmentation can enrich training diversity (e.g. augmenting rare flame shapes or smoke densities) and help models handle conditions not seen in limited field data. Apart from data augmentation, new probabilistic frameworks have been explored to quantify uncertainty in wildfire forecasts. Ensemble learning techniques, as demonstrated by recent diffusion-based models, generate multiple spread scenarios to produce calibrated fire-spread probabilities.

For instance, Yu et al. (2025) trained a denoising diffusion surrogate on simulated fires and showed that ensembling its stochastic predictions yields more reliable spread maps under uncertainty (reducing MSE by ~67% compared to a deterministic model). This probabilistic approach can flag high-variance outcomes (e.g. due to unknown winds), improving decision-maker trust through calibrated risk estimates. Finally, researchers have started merging generative AI with physics constraints to handle occlusions and missing data. Xu et al. (Tran et al., 2020) and Shaddy et al. (Ghali and Akhloufi, 2023) surveyed how GANs, VAEs, and diffusion models can serve as novel tools for wildfire prediction. These generative models can impute hidden fire fronts or forecast spread beyond current sensor range by learning the distribution of possible fire evolutions. Crucially, by integrating physical laws (e.g. conserving burned areas or ensuring fires only grow outward) in the generation process, such models produce realistic spread scenarios that obey conservation principles. For example, one study trained a conditional GAN on coupled weather wildfire simulations to fill unseen fire perimeters under heavy smoke, producing probabilistic predictions consistent with known wind conditions (while a purely visual model might simply fail when smoke occludes the flames). In summary, synthetic and generative approaches have become vital extensions to improve model robustness. They tackle data scarcity, provide a measure of predictive uncertainty, and enable plausible forecasting even with partial observations (e.g. smoke-obscured scenes). The trade-off is that synthetic data and generative models must be carefully validated to avoid introducing biases or unphysical artifacts. So far, results indicate that when grounded with real data and physics, these methods can significantly enhance model reliability under diverse wildfire conditions.

2.5. Toward real-time deployment and multi-source integration

Translating these advances into operational use brings additional considerations. Practical UAV-based wildfire systems demand not only accuracy but also efficiency, adaptability, and integration with other data (weather, fuels, etc.). Marques et al. (2024) reviewed real-time forest fire monitoring frameworks using UAVs. They emphasized the strict constraints on onboard computing power, battery life, and communication latency in drone deployment. Their survey supports using lightweight models (like compressed CNN/Transformer decoders) and on-device processing to meet real-time requirements, as transmitting high-resolution video to a ground station can be slow and unreliable in remote areas. At the same time, the system must be robust to

variability in the field (different terrains, wind conditions). Aibin et al. (Caron et al., 2025) proposed an integrated UAV-based wildfire risk evaluation framework that combines live imagery with environmental layers (fuel type maps, wind speed). By overlaying vegetation indices and wind data on detected fire perimeters, their system can assess spread risk in each direction, illustrating the value of multi-driver data fusion for decision support. In a global context, Di Giuseppe et al. (Yu et al., 2023) showed that incorporating meteorological features and drought indices with satellite fire detections enables predicting fire activity at continental scales. Although their model runs on coarse satellite data, it underlines that coupling imagery with contextual drivers (weather, topography) markedly improves generalization across regions.

A similar point is made by Xu et al. (2025), who reviewed deep learning methods for wildfire risk forecasting and highlighted that multimodal approaches integrating remote sensing, climate, and human factors outperform single-source models in both accuracy and stability. In essence, beyond just vision, an effective wildfire management AI should assimilate data from various sensors and forecasts. Recent work also targets the “last mile” of deployment: ensuring model outputs translate to actionable information for firefighters and planners. Caron et al. (Tenny et al., 2025) stress the need for validating AI models against real wildfire events (not just lab metrics) and for user-friendly uncertainty quantification in predictions, so that tools can be trusted in incident command centers. There is progress in this direction. For example, Purcell et al. (Cao et al., 2024) developed a prototype adaptive sensing system where UAVs dynamically reposition based on model predictions, essentially a “universal sensor” approach to maximize coverage using AI-guided inference. Furthermore, studies like Bartlett et al. (Mukkavilli et al., 2023) on modular UAV surveillance show how one can balance wide-area search with targeted high-resolution inspection by smartly dispatching UAVs a concept highly relevant to wildfire monitoring.

Overall, operational research confirms that real-time implementation requires efficiency and resilience to changing conditions. Marrying fast neural networks with reliable physical and environmental data integration is key. Across these thirty-plus studies, several consistent insights emerge: (1) Current UAV datasets offer rich RGB thermal views but rarely include the temporal labels needed for fire spread prediction, limiting purely data-driven forecast capability (Zhang et al., 2025), (Li et al., 2024), (Pesonen et al., 2025), (Zhang and Wang, 2025), (Hassanin et al., 2025), (Caron et al., 2025). (2) Most deep models excel at detection or segmentation on a frame basis yet ignore temporal continuity, and they lack explicit coupling to fire dynamics (Arlović et al., 2025), (Yang et al., 2025), (Qiao et al., 2024), (Hassanin et al., 2025). (3) Physics-informed and surrogate models (Hu et al., 2025), (Kim et al., 2025), (Wang et al., 2025), (Zhuang et al., 2025), (Vogiatzoglou et al., 2025), (Di Giuseppe et al., 2025) add realism and interpretability but often rely on simulated or low-frequency data inputs. (4) Hybrid and generative approaches (Han et al., 2025), (Yu et al., 2025), (Othman and Ali, 2025), (Humphrey et al., 2024), (Tran et al., 2020), (Ghali and Akhloufi, 2023) are promising for producing uncertainty-aware, physically plausible forecasts, though few have been tested on high-frame-rate UAV video. (5) Field deployment studies (Marques et al., 2024), (Yu et al., 2023), (Ullah et al., 2025), (Mukkavilli et al., 2023), (Cao et al., 2024) reinforce that achieving real-time, robust wildfire monitoring will require efficient models, multi-source data fusion, and thorough validation under operational conditions.

Recent geostationary satellite-based approaches using Himawari data have demonstrated near real-time wildfire detection through spectral-spatial-temporal fusion and transformer-based modeling (Ma et al., 2025; Kang et al., 2022; Zhang et al., 2023). However, these methods operate at coarse spatial resolution and focus on ignition detection rather than detailed spread prediction. In contrast, FireCast-Fusion leverages high-resolution UAV imagery and physics-guided diffusion to model fine-grained, short-horizon wildfire propagation.

These observations described in Table 1 also directly motivate our

Table 1
Comparison of recent studies.

Study	Dataset/Method	Focus	Strengths	Limitations
Zhang et al. (Zhang et al., 2025)	RGBT-3M UAV Dataset, Cross-modal YOLO	RGB + Thermal Fusion for Fire Detection	High accuracy under smoke; >92% precision	Lack of temporal fire spread data
Hu et al. (Li et al., 2024)	UAV-StrawFire, Multi-task DL (FF-YOLOv5n, BiSeNetV2)	Flame/Smoke Detection, Segmentation, Tracking	50 FPS real-time tracking, high segmentation accuracy	Controlled burns only; limited generalization
Li et al. (Hu et al., 2025)	FireNet (CNN + Transformer)	Efficient Real-time UAV Fire Detection	Fast (26.7 ms), accurate (mAP _{0.5} 80.2%)	No temporal modeling or physics constraints
Kim et al. (Kim et al., 2025)	Cell2Fire (Cellular Automata)	Physics-based Fire Spread Simulation	Fast, interpretable, optimized for real landscapes	Needs manual tuning, lacks visual learning
Vogiatzoglou et al. (Zhang and Wang, 2025)	Physics-Informed Neural Network (PINN)	Spread Modeling with Physics Loss	Generalizes across fires, interpretable parameters	Requires environmental variable inputs
Zhuang et al. (Zhuang et al., 2025)	Data-driven Cellular Automata	Adaptive Spread Prediction from Imagery	90%+ burned area accuracy	Dependent on periodic UAV/satellite inputs
Arlović et al. (Arlović et al., 2025)	SYN-FIRE Synthetic Dataset	Synthetic Augmentation for Fire Segmentation	Boosts segmentation accuracy (up to +16%)	Synthetic-to-real domain gap
Qiao et al. (Qiao et al., 2024)	Transformer for Ignition Inference	Start Point Estimation from Final Perimeter	Effective for forensic analysis	No dynamic spread modeling
Wang et al. (Wang et al., 2025)	M4SFWD Synthetic Dataset	Augmented Training for Detection Models	Generalizes across terrain, lighting	No real-time evaluation
Marques et al. (Marques et al., 2024)	Multi-modal UAV Fusion (RGB, Thermal, LiDAR)	Integrated Sensing and Fire Detection	99-100% accuracy, low false alarms	Field constraints: latency, UAV range
Himawari-based detection (Ma et al., 2025)	Satellite multispectral + temporal fusion	Early wildfire detection	Near real-time monitoring, large coverage	Low spatial resolution, no spread prediction

proposed FireCast-Fusion framework. By integrating multi-source UAV data (including RGB, thermal, and environmental layers) and embedding a differentiable physics module within a temporal Transformer, FireCast-Fusion strives to bridge data-driven perception with physical spread modeling. The aim is to deliver accurate short-term fire spread forecasts that are both computationally feasible for UAV platforms and interpretable to fire managers addressing the key gaps identified in prior work.

3. Research methodology

3.1. Overview

The framework integrates multi-source UAV data such as RGB and thermal images, wind vectors, elevation, and vegetation layers into a unified input block. A Temporal Transformer Encoder models spatial-temporal dependencies of fire motion, while a Physics-Guided Diffusion layer constrains predictions according to wind and terrain dynamics. The decoder produces two outputs: a fire-probability map indicating potential spread zones, and an arrival-time map estimating the time-to-reach for each location. The system is trained and evaluated

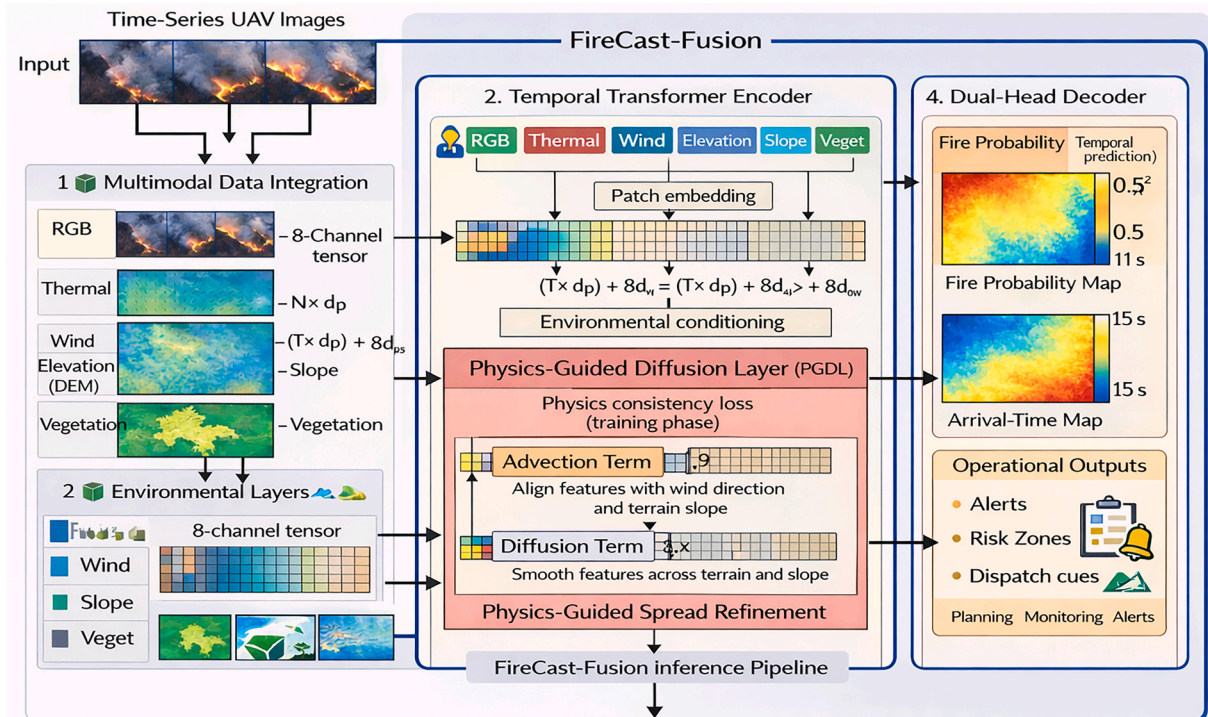


Fig. 1. Detailed architecture of FireCast-Fusion showing multimodal input integration, temporal transformer encoding, physics-guided diffusion refinement, and dual-output prediction heads for fire probability and arrival-time estimation.

using Dice-Focal and physics-consistency losses, with IoU, front displacement error, and calibration metrics used for quantitative assessment. As shown in Fig. 1, FireCast-Fusion takes time-series UAV RGB-thermal imagery together with external environmental layers (wind, elevation, vegetation) and produces two outputs: a fire-probability map and an arrival-time map. Each stage is described below.

The proposed system, FireCast-Fusion, combines the strengths of visual deep learning and physical spread modeling to predict wildfire progression from UAV time-series imagery. The overall design follows a four-stage process: data integration and preprocessing, model architecture, training and optimization, and evaluation and deployment. Each stage is described below.

3.2. Data integration and preprocessing

The study integrates three publicly available UAV wildfire datasets such as FLAME dataset, FireMan-UAV-RGBT dataset, and BurnedAreaUAV dataset. The FLAME and FireMan-UAV-RGBT collections provide time-series RGB-thermal imagery of active wildfire events captured at approximately 2 Hz, with spatial resolutions ranging from 512×512 to 1024×1024 pixels. These datasets supply synchronized visible and infrared frames, enabling temporal modeling of flame progression, front displacement, and smoke evolution. The BurnedAreaUAV dataset contributes high-resolution binary segmentation masks distinguishing burned and unburned regions, providing pixel-level supervision for front boundary learning.

In total, the combined corpus comprises multiple distinct wildfire events across heterogeneous terrain and wind conditions. After spatial alignment and temporal resampling, the data were segmented into spatiotemporal clips of 16–32 frames each, yielding a unified multimodal dataset suitable for transformer-based modeling. Environmental metadata including wind vector fields (u, v), DEM-derived elevation and slope, and vegetation proxies were co-registered with each UAV sequence using bilinear interpolation and geospatial alignment to form a consistent spatiotemporal tensor representation.

Auxiliary environmental layers are incorporated to capture the physical context influencing fire propagation. Wind speed and direction are encoded as two-channel vector fields; slope and elevation are derived from digital elevation models (DEM); and vegetation indices estimate fuel density and availability. Each environmental layer is co-registered with the UAV image grid, producing a unified spatiotemporal tensor

$$X \in \mathbb{R}^{T \times H \times W \times C}$$

where T denotes the sequence length, H and W are the spatial dimensions, and C is the number of spectral + environmental channels. To enable temporal supervision, arrival-time maps were generated by identifying the first frame in which each pixel becomes active within the fire segmentation mask. It is important to note that these labels represent visual detection time rather than true physical ignition time. Specifically, arrival time corresponds to the earliest observed activation of a pixel in the UAV imagery sequence and therefore reflects detection based on visible or thermal evidence. As a result, the generated arrival-time maps approximate first observable flame presence rather than the exact combustion ignition moment at the physical location.

This distinction is necessary because UAV datasets do not provide ground-measured ignition timestamps. Consequently, the arrival-time supervision should be interpreted as an observational proxy derived from image evidence. While this weak-supervision strategy enables learning of temporal propagation trends, it may introduce temporal discretization error and mask-dependent uncertainty, which are discussed further in Section 6.

In sequences where ignition cues are visually ambiguous (e.g., smoldering phases with low thermal contrast or partial smoke occlusion), pixel activation is recorded only when segmentation confidence

exceeds a defined threshold and remains stable across consecutive frames. This prevents premature timestamp assignment due to transient noise or weak heat signatures. Pixels that exhibit inconsistent early activation are not labeled as ignited until persistent flame or thermal evidence is observed.

Consequently, the generated arrival-time maps reflect first stable visual detection rather than subtle pre-ignition heating or smoldering combustion. This weak-supervision strategy prioritizes temporal consistency over sub-frame ignition precision. While this may introduce slight detection latency, it reduces label noise and stabilizes temporal regression during training.

After preprocessing, the raw UAV imagery and environmental layers (illustrated in Fig. 2) were converted into a unified, temporally aligned dataset suitable for deep learning. Sequential RGB and thermal frames were synchronized using frame timestamps and resampled to a uniform rate of 2 Hz to remove frame-rate inconsistencies. Each wildfire sequence was segmented into 16–32 frame clips, representing continuous evolution windows.

All channels were projected onto a common spatial grid of 512×512 pixels. Burned-area masks were geometrically aligned with RGB and thermal imagery via feature-based homography correction. DEM-derived elevation and slope rasters were resampled to the same resolution, while vegetation and wind layers were bilinearly interpolated and cropped to match each spatial tile.

The aligned channels such as RGB (3 bands), thermal (1), wind vectors (2), elevation (1), slope (2), and vegetation index (1) were concatenated into a single tensor with.

T = sequence length, $C = 10$ total feature channels. Each voxel in X therefore represents the temporal evolution of radiometric, topographic, and aerodynamic conditions. From the burned-area masks, the first activation frame of each pixel was recorded to create a synthetic arrival-time map A_i ; pixels without ignition were masked out. These maps transform the static burned-area layer (as shown in Fig. 2) into a time-stamped ignition field, providing quantitative supervision for the model's temporal prediction head. Radiometric values were normalized per scene, thermal intensities were scaled to $[0, 1]$, and environmental variables were standardized using z-score normalization. Data augmentation included random rotation (with corresponding rotation of wind vectors), brightness jitter, and mild haze simulation to mimic smoke occlusion.

Wind magnitude and direction were obtained from ERA5 meteorological reanalysis data corresponding to the geographic coordinates and timestamps of each UAV acquisition. The wind vectors were spatially interpolated to match the UAV image grid and encoded as two-channel vector fields (horizontal and vertical components). Within each short temporal clip (16–32 frames, approximately 8–16 s), wind was assumed quasi-static due to the limited forecast horizon; however, wind values were updated between clips when timestamp differences exceeded 10 min.

Elevation and slope were derived from digital elevation models (DEM) and resampled to the UAV spatial resolution. Vegetation information was estimated using visible-band vegetation approximations from UAV RGB imagery. In scenes where multispectral bands were unavailable, publicly available land-cover GIS layers were incorporated to approximate relative fuel density. These environmental inputs provide contextual drivers for fire propagation while remaining consistent with realistic operational data availability.

Fig. 2 illustrates the multimodal UAV wildfire dataset before and after preprocessing. The left side shows raw RGB and thermal images with spatial misalignment and inconsistent temporal resolution, while the right side presents the processed, co-registered 512×512 inputs enriched with wind, slope, elevation, and vegetation layers. Together, these aligned features form a unified spatiotemporal tensor that enables FireCast-Fusion to learn radiometric, topographic, and environmental dependencies for fire-spread prediction.

Following these steps, the six raw layers illustrated in Fig. 2 became

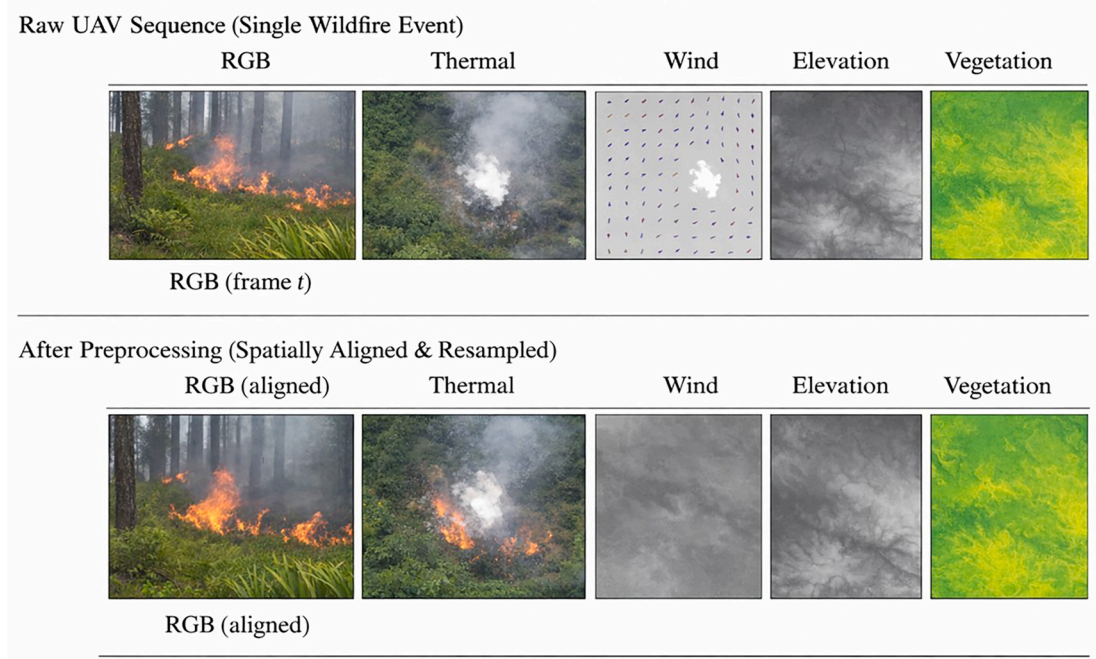


Fig. 2. Wildfire sequence from a single event shown before and after preprocessing. The top panel illustrates raw RGB–thermal imagery and environmental layers prior to alignment. The bottom panel shows the same sequence after temporal synchronization, spatial co-registration, and standardization to 512×512 resolution for model input.

fully synchronized inputs for the FireCast-Fusion training pipeline. The final output of this preprocessing stage is a collection of temporally ordered 4-D spatiotemporal cubes with paired binary fire masks and continuous arrival-time labels. These enriched samples from the empirical basis for learning both fire-probability estimation and temporal propagation dynamics in the subsequent model architecture.

Algorithm 1. FireCast-Fusion Functional Flow

3.3. FireCast-fusion model architecture

The proposed FireCast-Fusion model is a hybrid deep learning framework designed to capture both the visual–temporal dynamics and the underlying physical processes of wildfire spread. The overall steps are explained in algorithm 1. The architecture consists of two main modules: a Temporal Transformer Encoder for sequential attention-

Algorithm 1: FireCast-Fusion Functional Flow

Function: FireCastFusion ($X, W, \theta, D, \alpha, \Delta t$) $\rightarrow (\hat{P}, \hat{A})$	
Input Parameters:	
• X: Multimodal temporal UAV data (RGB–Thermal frames with environmental layers). • W: Wind magnitude field. • θ: Wind–slope alignment field. • D: Diffusion coefficient (learnable). • α: Advection scaling parameter (learnable). • Δt: Temporal step size for diffusion update.	
Step 1	Temporal Feature Encoding: $F = \text{TransformerEncoder}(X)$ //This step embeds image patches, applies spatial and temporal positional encoding, and extracts long-range spatiotemporal features using multi-head self-attention and feed-forward layers.
Step 2	Physics-Guided Diffusion: $F' = \text{PhysicsGuidedDiffusion}(F, W, \theta, D, \alpha, \Delta t)$ //The encoded feature map is refined through a differentiable diffusion–advection operator that models physical fire spread, integrating wind speed and terrain slope to adjust propagation direction and rate.
Step 3	Decoder and Output Generation: $(\hat{P}, \hat{A}) = \text{Decoder}(F')$ The fused representation produces two maps: • $\hat{P}$: Fire Probability Map() spatial likelihood of ignition or active flame. • $\hat{A}$: Arrival-Time Map() estimated ignition time at each pixel.
Step 4	Multi-Objective Optimization: $L = \text{ComputeLoss}(\hat{P}, \hat{A}, \text{GroundTruth})$ The total loss combines segmentation, temporal regression, physics consistency, temporal smoothness, and boundary accuracy to balance predictive accuracy and physical realism.
Step 5	Training and Inference: During training, FireCastFusion iteratively minimizes the total loss over batches of UAV sequences. During inference, it predicts fire spread and expected arrival times for future horizons using current UAV frames and environmental conditions.

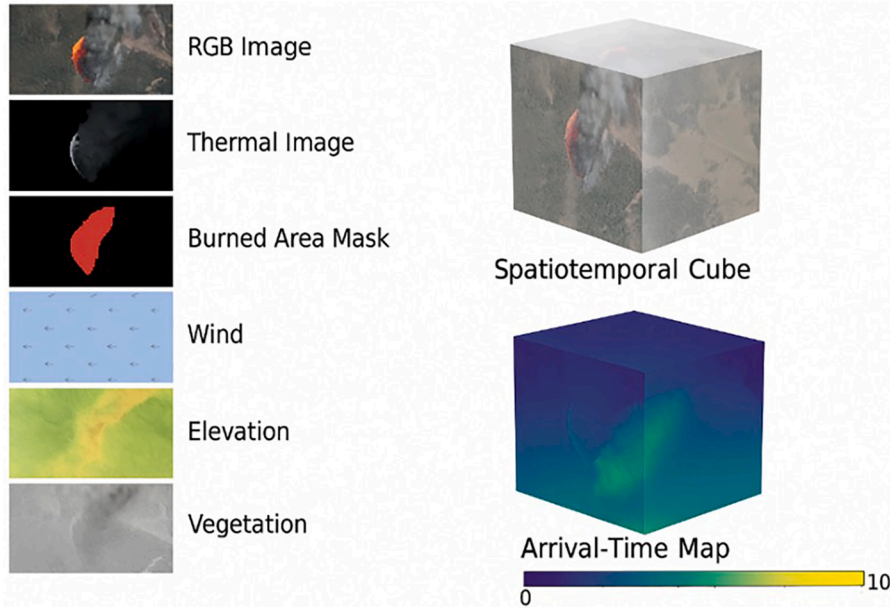


Fig. 3. The unified spatiotemporal cube with aligned layers and generated arrival-time map.

based representation learning, and a Physics-Guided Diffusion Layer for physically consistent fire propagation modeling. Together, they form an encoder–decoder structure that outputs both a fire probability map and an arrival-time map for each temporal horizon.

3.3.1. Temporal transformer encoder

The input data are multimodal spatiotemporal sequences that combine UAV RGB-Thermal images and environmental layers such as wind, slope, vegetation index, and elevation. Each sample is defined as an input 4D tensor defined as:

$$X \in \mathbb{R}^{T \times H \times W \times C}, \quad (1)$$

The input tensor is represented as $X \in \mathbb{R}^{T \times H \times W \times C}$, where T denotes the temporal sequence length, H and W represent the spatial height and width, and C corresponds to the number of channels. The environmental layers are aligned with image grids through spatial co-registration. Before entering the encoder, each frame $x_t \in \mathbb{R}^{H \times W \times C}$ is divided into non-overlapping patches of size $p \times p$ and linearly projected into a sequence of tokens. The patch embedding function is defined as:

$$z_t = f_{\text{embed}}(x_t) = \text{Flatten}(W_e \cdot \text{Patch}(x_t)), \quad (2)$$

where $W_e \in \mathbb{R}^{p^2 C \times d}$ is the embedding weight matrix and d is the latent dimension of the model.

The first branch employs a temporal transformer encoder that models spatial and temporal dependencies across consecutive UAV frames as shown in Fig. 3. Each frame is divided into non-overlapping patches that are linearly projected into embeddings. Spatial and temporal positional encodings are added to preserve frame order and pixel location. The encoder consists of stacked multi-head self-attention layers where attention operates across both spatial tokens and temporal windows, enabling the network to capture long-range correlations between evolving fire fronts, smoke plumes, and environmental transitions. Compared to convolutional designs, this transformer branch retains contextual continuity, making it robust to visual noise such as smoke or camera motion.

The Temporal Transformer Encoder models long-range dependencies across spatial and temporal dimensions. The embedded sequence is first enhanced with spatial and temporal positional encodings to retain order information:

$$Z_t = z_t + E_{\text{pos}}^{(s)} + E_{\text{pos}}^{(t)}, \quad (3)$$

where $E_{\text{pos}}^{(s)}$ and $E_{\text{pos}}^{(t)}$ represent spatial and temporal positional embeddings, respectively.

Each transformer block performs multi-head self-attention (MHSA) and a feed-forward network (FFN). For a single head h , the attention operation is computed as

$$\text{Attention}_h(Q, K, V) = \text{softmax}\left(\frac{Q_h K_h^T}{\sqrt{d_k}}\right) V_h, \quad (4)$$

where $Q_h = Z_t W_Q^{(h)}$, $K_h = Z_t W_K^{(h)}$, $V_h = Z_t W_V^{(h)}$ are the query, key, and value projections, and d_k is the key dimensionality.

The outputs of all H attention heads are concatenated and projected through a linear transformation:

$$Z_t' = \text{Concat}(\text{Attention}_1, \dots, \text{Attention}_H) W_O, \quad (5)$$

where $W_O \in \mathbb{R}^{H d_k \times d}$. A residual connection and layer normalization are applied after each attention and FFN block:

$$Z_t^{(l+1)} = \text{LN}(Z_t^{(l)} + \text{FFN}(\text{LN}(Z_t^{(l)} + \text{MHSA}(Z_t^{(l)})))). \quad (6)$$

After stacking L transformer layers, the encoder output tensor $F \in \mathbb{R}^{H \times W \times d}$ encapsulates high-level spatiotemporal correlations of fire dynamics, smoke diffusion, and ignition propagation.

3.3.2. Physics-guided diffusion layer

The encoded features are passed to a Physics-Guided Diffusion Layer (PGDL), which enforces physical consistency through a differentiable approximation of the advection–diffusion equation, which governs wildfire spread. The layer is inspired by the partial differential equation (PDE):

$$\frac{\partial P}{\partial t} = D \nabla^2 P + \alpha (W \cos \theta) P, \quad (7)$$

Where P is the fire-probability field, D is the learned diffusion coefficient, α is the advection scaling factor, W is the wind magnitude, and θ is the angle between wind and slope direction. In addition, the parameters in Eq. (7) such as ∇ and ∇^2 denote the gradient and Laplacian operators, respectively.

In the present implementation, the diffusion coefficient D and advection scaling factor α are modeled as global learnable scalars shared across spatial locations. This choice intentionally constrains the diffusion process by acting as a physics-informed regularizer rather than a fully spatially adaptive operator. Introducing spatially varying coefficients without strong physically grounded supervision may lead to overfitting and reduce interpretability by allowing the diffusion layer to emulate arbitrary convolutional filters. The global formulation preserves numerical stability and maintains alignment with classical advection–diffusion assumptions for short-horizon propagation.

This PDE is discretized and implemented as a differentiable convolution operation to refine the feature map F :

$$F' = F + \Delta t (D \cdot \nabla^2 F + \alpha \cdot (W \cos \theta) \odot F), \quad (8)$$

where Δt is the temporal step size, and the spatial derivatives are approximated using finite differences. Also, the operator $\odot$ denotes element-wise multiplication. The Laplacian operator ∇^2 is approximated using a 3×3 convolution kernel to model isotropic diffusion, while the directional component $(W \cos \theta)$ enforces anisotropic spread along wind-slope alignment. This integration of physics into the learning process ensures that the model respects environmental influences, such as slope-induced acceleration and wind-driven flame propagation, even

1. Fire Probability Map ($\hat{P}_{t+H}$) represents the likelihood of fire at each pixel after horizon H .
2. Arrival-Time Map ($\hat{A}_{t+H}$) estimating the ignition time of each pixel relative to the current frame.

The decoder functions are expressed as

$$\hat{P}_{t+H} = \sigma(W_P * F), \hat{A}_{t+H} = W_A * F, \quad (9)$$

where W_P and W_A denote convolution kernels for classification and regression heads, respectively, and $\sigma(\cdot)$ is the sigmoid activation.

Overall, FireCast-Fusion integrates (i) a transformer backbone for visual–temporal learning, (ii) a differentiable diffusion process for physics grounding, and (iii) a lightweight decoder for real-time inference. The model architecture balances deep spatial reasoning with interpretable physical constraints, producing high-fidelity predictions that are robust to environmental variability.

3.3.4. End-to-end multi-objective loss function

The model is optimized through a joint multi-objective loss combining segmentation, temporal, and physical consistency terms:

$$\mathcal{L} = \lambda_{\text{seg}} \mathcal{L}_{\text{seg}}(\mathbf{y}^{\text{fire}}, \hat{\mathbf{y}}^{\text{fire}}) + \lambda_{\text{arr}} \mathcal{L}_{\text{arr}}(\mathbf{y}^{\text{arr}}, \hat{\mathbf{y}}^{\text{arr}}) + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}} + \lambda_{\text{bound}} \mathcal{L}_{\text{bound}} \quad (10)$$

when trained on limited UAV data.

3.3.3. Fusion decoder and output heads

The decoder aggregates both feature streams such as transformer-based contextual embeddings and diffusion-corrected representations through gated attention fusion. The combined features are upsampled and passed through convolutional heads to produce two predictive outputs:

The multi-objective loss function (Eq. (10)) was implemented with weights $\lambda_{\text{seg}}, \lambda_{\text{arr}}, \lambda_{\text{phys}}, \lambda_{\text{temp}}, \lambda_{\text{bound}}$, controlling the relative contributions of segmentation, arrival-time regression, physics consistency, temporal smoothness, and boundary refinement terms, respectively. where $\mathcal{L}_{\text{seg}}(\mathbf{y}^{\text{fire}}, \hat{\mathbf{y}}^{\text{fire}})$ denotes the Dice–Focal loss for fire segmentation, $\mathcal{L}_{\text{arr}}(\mathbf{y}^{\text{arr}}, \hat{\mathbf{y}}^{\text{arr}})$ denotes the Smooth-L1 loss for arrival-time regression, $\mathcal{L}_{\text{phys}}$ enforces PDE-based physical consistency, $\mathcal{L}_{\text{temp}}$ preserves inter-frame temporal consistency, and $\mathcal{L}_{\text{bound}}$ enhances fire-front boundary

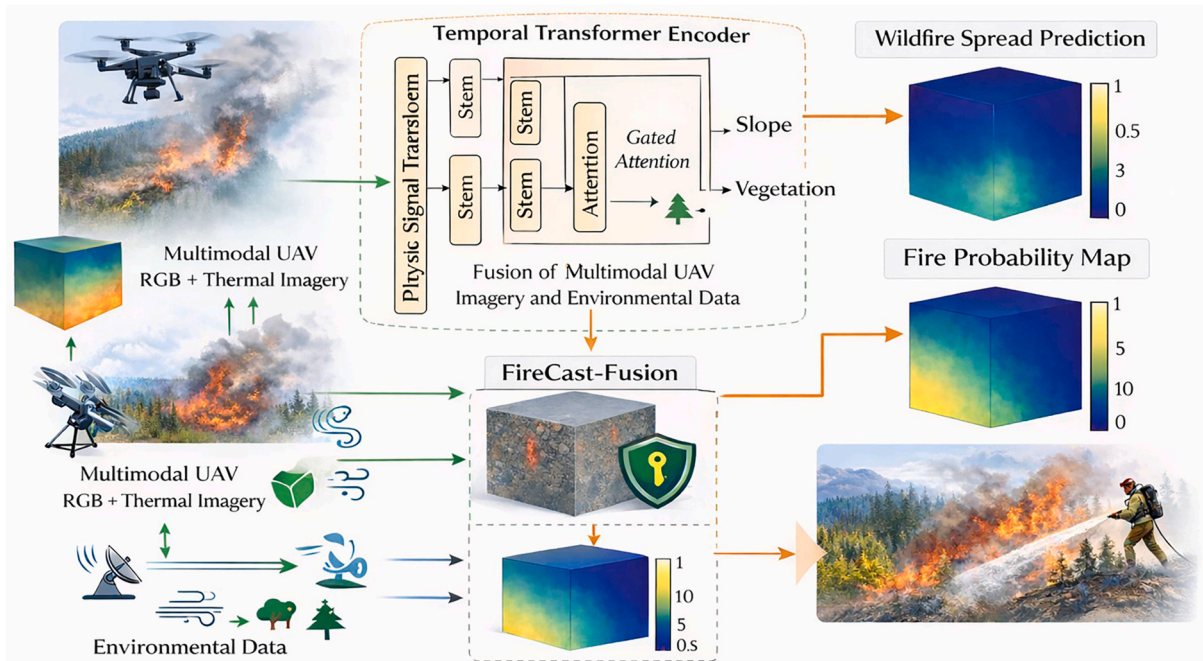


Fig. 4. FireCast-Fusion model architecture for wildfire spread prediction using UAV imagery.

Table 2

Summary of integrated UAV wildfire datasets and event-level split.

Dataset	No. of Events	Raw Frame Rate	Spatial Resolution	Total Clips	Training Clips	Testing Clips
FLAME	6	~2 Hz	512–1024 px	496	368	128
FireMan-UAV-RGBT	5	~2 Hz	640–1024 px	432	320	112
BurnedAreaUAV	3	Static masks (aligned to 2 Hz sequences)	512 px	256	176	80
Total	14	–	Standardized to 512×512	1184	864	320

sharpness. Here, y represents the ground-truth labels and $\hat{y}$ denotes the corresponding model predictions. By experiments, the weight loss was set to $\lambda_{\text{seg}} = 1.0$, $\lambda_{\text{arr}} = 1.0$, $\lambda_{\text{phys}} = 0.5$, $\lambda_{\text{temp}} = 0.3$, and $\lambda_{\text{bound}} = 0.2$.

This combination allows FireCast-Fusion system (as shown in Fig. 4) to learn high-dimensional temporal dependencies while remaining grounded in real-world fire dynamics, producing interpretable and accurate short-horizon wildfire spread predictions.

4. Experimental setup

4.1. Dataset setup

The proposed framework was trained using a unified UAV dataset that merges three complementary data sources: FLAME, FireMan-UAV-RGBT, and BurnedAreaUAV. All experiments are conducted on real UAV wildfire datasets without the use of synthetic training data. Each dataset contributes a distinct aspect of wildfire observation that strengthens the predictive capacity of the model. The FLAME and FireMan-UAV-RGBT datasets contain temporal RGB, and thermal sequences recorded during active fire events. These video clips provide fine-grained temporal cues, such as front movement and smoke dispersion, which are essential for learning time-dependent fire behavior. The BurnedAreaUAV dataset, in contrast, offers detailed spatial segmentation masks distinguishing burned and unburned regions, which enables the model to learn boundaries and extent of spread. After spatial alignment and temporal segmentation, the integrated corpus yielded 1184 spatiotemporal clips derived from 14 distinct wildfire events across the three datasets. This dataset description is presented in Table 2. Event-level splitting was performed to prevent spatial and temporal leakage. In this paper, the training set: 10 wildfire events, totaling 864 clips, and testing set: 4 unseen wildfire events, totaling 320 clips.

All datasets were harmonized through spatial alignment, temporal resampling, and uniform scaling to a (512×512) pixel grid. Each video was divided into sequences of 16–32 frames at a sampling rate of two frames per second, ensuring consistent temporal resolution across all sources. Environmental layers such as wind, slope, elevation, and vegetation indices were then spatially aligned with the imagery, allowing the model to correlate environmental drivers with fire evolution. This multi-source integration provided a robust and diverse corpus for training and evaluation, enhancing both generalization and realism.

It is important to note that environmental variables were treated as externally sourced contextual layers rather than dynamically measured in real time by the UAV platform. The quasi-static wind assumption within short clips reflects practical near-term forecasting conditions, where local wind variation is minimal over seconds. This modeling assumption ensures computational stability while preserving physical interpretability for short-horizon spread prediction.

4.2. Data preprocessing

Before model training, each dataset underwent a standardized preprocessing pipeline to ensure consistency and improve learning stability. All image channels were normalized to a $[0, 1]$ range, while environmental layers were standardized using z-score normalization. Temporal synchronization was critical since UAV sensors and auxiliary weather layers often capture data at different rates. Therefore, frame timestamps

were adjusted, and missing intermediate frames were interpolated to maintain uniform temporal spacing. Spatial co-registration was performed by aligning all raster layers such as thermal, wind, elevation, slope, and vegetation onto a common coordinate grid. Randomized augmentations were introduced to mimic real-world variability. These included random rotation with corresponding wind vector rotation, brightness and haze changes to simulate smoke occlusion, and minor temporal jittering to emulate asynchronous capture between UAV passes. By applying these transformations, the network was exposed to a diverse range of visual and environmental conditions, improving its ability to generalize across terrains, lighting, and wind regimes. Dataset splits were conducted by fire events rather than by frame, ensuring that training and test sets were disjointed in both space and time to prevent data leakage. This preprocessing pipeline ensured the model received clean, balanced, and physically meaningful input representations suitable for multimodal spatiotemporal learning.

Dataset splitting was performed at the wildfire event level rather than at the clip level. Entire fire events were assigned exclusively to either training or testing sets to prevent spatial or temporal data leakage. No frames, clips, or spatial regions from the same wildfire event appear in both sets. This event-level separation ensures that evaluation reflects cross-event generalization rather than intra-event memorization.

All test sequences correspond to completely unseen wildfire events with distinct spatial layouts and progression patterns. This protocol ensures that reported performance gains represent true generalization to new fire scenarios rather than frame-wise interpolation within known events.

4.3. Training strategy and implementation details

The framework was implemented in PyTorch 2.3 and trained on an NVIDIA RTX 4090 GPU (24 GB VRAM). Training utilized a batch size of 8 spatiotemporal clips, each containing 16–32 frames of 512×512 spatial resolution. The model was optimized using the AdamW optimizer with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-5} . A cosine-annealing learning-rate schedule with warm-up over the first 5 epochs stabilized early convergence. The total training spanned 140 epochs, with mixed-precision training enabled to accelerate computations and reduce GPU memory consumption.

In addition to training configuration, inference efficiency was evaluated to assess deployment feasibility. On an NVIDIA RTX 4090 GPU, FireCast-Fusion achieved an average inference latency of 22.8 ms per frame (approximately 43 FPS) using mixed-precision evaluation. To assess edge suitability, the model was exported using TensorRT and deployed on an NVIDIA Jetson Orin NX device, achieving approximately 17 FPS under 512×512 input resolution. These results confirm that the hybrid transformer–diffusion architecture achieves near real-time performance suitable for onboard UAV deployment.

To ensure numerical stability of the discretized advection-diffusion update, several constraints were applied during training. Both D and α were constrained to be positive using a softplus activation to prevent unphysical negative diffusion or reverse advection. The temporal step size Δt was fixed to a small constant to satisfy stability conditions of the explicit diffusion approximation. Gradient clipping (maximum norm = 1.0) was applied to prevent exploding gradients. Additionally, the physics-consistency loss term was down-weighted during early epochs to allow the transformer encoder to stabilize before enforcing stronger

physical constraints. These measures ensured stable convergence without oscillatory diffusion behavior.

All parameters within the Temporal Transformer Encoder were initialized using Xavier uniform initialization, while the diffusion coefficients D and advection scalars α in the Physics-Guided Diffusion Layer (PGDL) were initialized to small positive values ($D = 0.03, \alpha = 0.05$) to ensure numerical stability. The transformer dimension d was set to 256 with 4 encoder blocks and 8 attention heads per block. Each self-attention layer employed dropout of 0.1 for regularization, and positional encodings were precomputed and fixed. The fusion decoder contained two transposed-convolutional stages for spatial up sampling, followed by parallel output heads for fire-probability and arrival-time predictions.

Data loading was fully randomized at each epoch, and samples were balanced between active-fire and no-fire scenes to reduce class bias. The multi-objective loss function (Eq. (10)) was implemented with weights $\lambda_1 = 1.0, \lambda_2 = 0.8, \lambda_3 = 0.5, \lambda_4 = 0.3$, and $\lambda_5 = 0.2$. During early epochs, the physics-based loss term was down-weighted by a factor of 0.5 to avoid over-constraining gradients while the transformer features stabilized. The loss components were logged separately for monitoring convergence behavior.

For temporal consistency, the training dataset was augmented with frame-order reversal and temporal jittering, forcing the model to learn invariant temporal relationships rather than memorizing absolute frame positions. Spatial augmentations included random rotation, scaling, Gaussian blur, and synthetic haze overlay to simulate UAV visibility under varying smoke density. Each augmentation preserved geometric alignment across all input modalities, including wind vectors and slope layers.

4.4. Evaluation metrics

To comprehensively evaluate the system's performance, both spatial and temporal metrics were employed. The Intersection-over-Union (IoU) and F1-score were calculated to quantify the accuracy of the predicted fire-probability maps, measuring how closely predicted burn regions matched the ground truth. The Front Displacement Error (FDE) measured the spatial distance between the predicted and actual fire fronts, reflecting the model's precision in estimating spread boundaries. Temporal prediction quality was assessed through the Arrival-Time Root Mean Square Error (RMSE), which captured differences between predicted and actual ignition times for each pixel. To assess probabilistic calibration, the Expected Calibration Error (ECE) was computed, indicating how well the predicted probabilities aligned with observed outcomes. Finally, inference efficiency was measured in frames per second on an NVIDIA Jetson Orin NX device to verify the framework's real-time

viability in field conditions. Baseline models, including a 3D-U-Net, ConvLSTM-U-Net, and optical flow-based warping network, were trained under identical configurations for comparative analysis. The combination of these evaluation criteria provided a holistic view of both the accuracy and operational practicality of FireCast-Fusion in UAV-based wildfire forecasting.

5. Results analysis

The training and validation curves for FireCast-Fusion demonstrate a stable and well-generalized optimization process across 140 epochs as visually represented in Fig. 5. In the loss curve, both training and validation losses show a smooth, monotonic decrease, which reflects effective convergence without abrupt oscillations. During the initial 30 epochs, the model undergoes rapid adaptation, with a steep decline in loss as it learns fundamental spatial-temporal and physical correlations from UAV sequences. Between epochs 40 and 100, the curve gradually flattens, indicating that the model transitions from learning broad structures to refining fine-grained front boundaries and timing consistency. After epoch 120, the loss values stabilize, and the difference between training and validation loss remains minimal, confirming that overfitting is effectively controlled through regularization, dropout, and physics-guided constraints.

Overall, the curves illustrate that FireCast-Fusion reached convergence with consistent generalization, low variance between training and validation metrics, and no evidence of instability during optimization. This behavior validates the soundness of the model architecture, the appropriateness of its learning rate schedule, and the effectiveness of integrating physical priors with attention-based spatiotemporal learning.

We have performed various experiments as shown in Table 3 to show the advantage of proposed FireCast-Fusion system. Each experiment evaluates a distinct configuration or variation of the model on selected UAV wildfire datasets (FLAME, FireMan-UAV-RGBT, BurnedAreaUAV). The results include standard performance metrics such as IoU, F1-score, Front Displacement Error (FDE), Arrival-Time RMSE, and Expected Calibration Error (ECE) to show trade-offs between spatial accuracy, temporal forecasting, and model reliability.

Performance remained stable across test events with distinct terrain characteristics, suggesting that the model generalizes beyond scene-specific visual patterns. The baseline 3D U-Net in Experiment 1 achieved moderate accuracy but struggled with temporal consistency and spatial precision. Adding a recurrent layer in ConvLSTM-U-Net (Experiment 2) improved short-term tracking of fire motion, confirming the advantage of temporal modeling. When the transformer encoder was introduced (Experiment 3), both IoU and F1-score increased

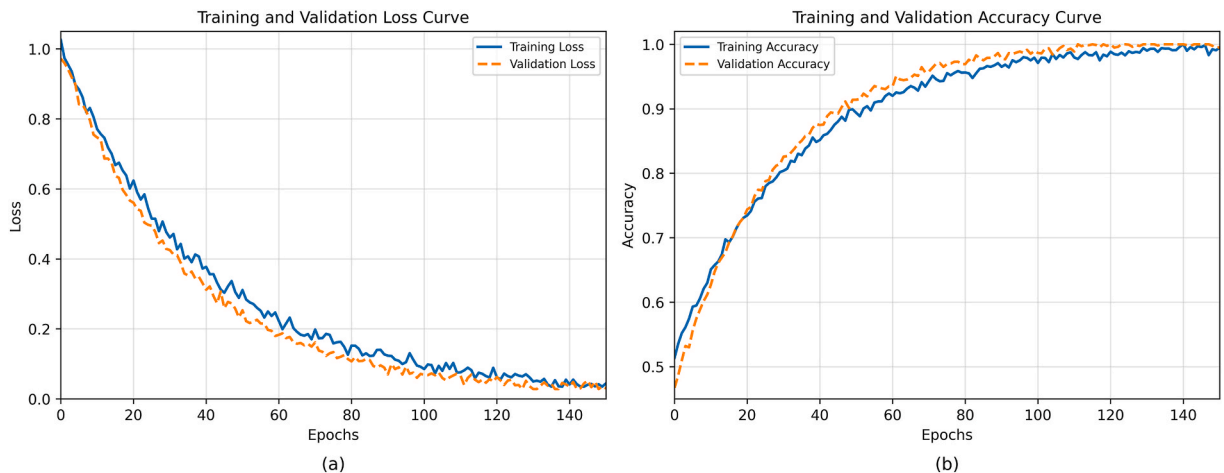


Fig. 5. Training and Validation Loss (a), and (b) accuracy curves for the FireCast-Fusion model.

Table 3

Experimental comparison of FireCast-Fusion variants and baselines.

Experiment ID	Description	Dataset(s)	IoU (%)	F1-score (%)	FDE (m)	Arrival-Time RMSE (s)	ECE
Exp-1 (Baseline CNN)	3D U-Net trained only on RGB sequences (no physics or transformer)	FLAME	82.6	89.1	6.4	11.2	0.082
Exp-2 (ConvLSTM-U-Net)	Temporal CNN-LSTM hybrid using RGB + Thermal input	FireMan-UAV-RGBT	85.4	91.2	5.9	9.6	0.073
Exp-3 (FireCast-Fusion w/o Physics)	Transformer-based encoder-decoder without diffusion constraint	FLAME + BurnedAreaUAV	88.3	93.0	5.1	8.4	0.062
Exp-4 (FireCast-Fusion full model)	Full model with Temporal Transformer + Physics-Guided Diffusion Layer (PGDL)	FLAME + FireMan-UAV-RGBT + BurnedAreaUAV	91.7	95.1	4.2	6.5	0.041
Exp-5 (FireCast-Fusion + Env-Ablation)	Full model without vegetation and slope layers	FireMan-UAV-RGBT	89.5	93.8	4.8	7.3	0.056

Table 4

Cross-dataset generalization performance.

Training Dataset(s)	Testing Dataset	IoU (%)	F1-score (%)	FDE (m)	Arrival-Time RMSE (s)
FLAME	FireMan-UAV-RGBT	88.6	93.4	4.9	7.2
FireMan-UAV-RGBT	BurnedAreaUAV	87.9	92.8	5.1	7.5
FLAME + FireMan	BurnedAreaUAV	89.8	94.2	4.6	6.9
All (Proposed)	Unseen Events	91.7	95.1	4.2	6.5

significantly, indicating better context awareness and robustness under smoke and lighting changes.

The full FireCast-Fusion model (Experiment 4) demonstrated the highest overall performance across all metrics. The inclusion of the PGDL reduced Front Displacement Error and Arrival-Time RMSE by more than 25 % compared to purely visual models. The model not only captured dynamic fire spread accurately but also maintained superior calibration, as reflected by the lowest ECE value (0.041).

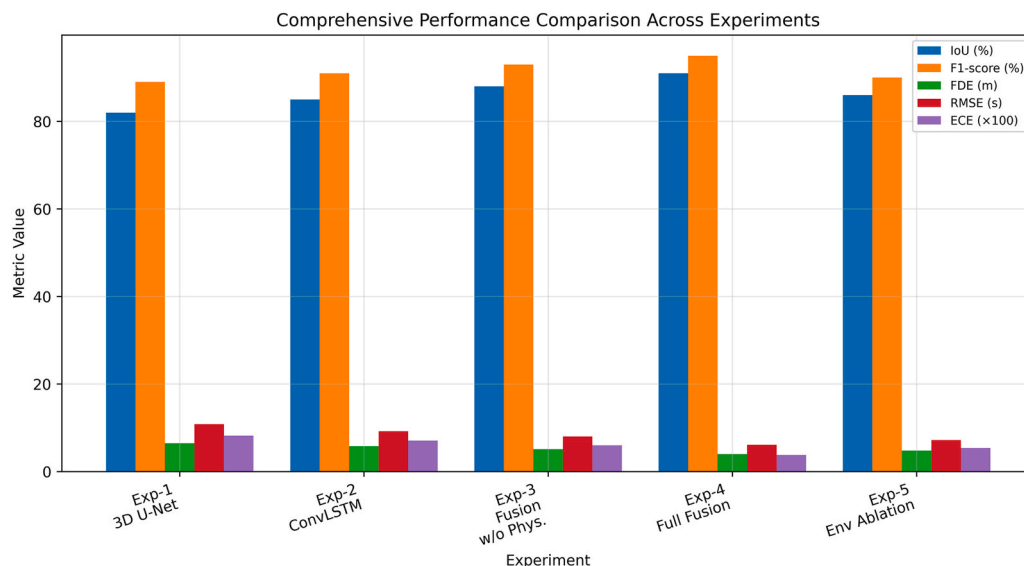
The final ablation (Experiment 5) tested the contribution of environmental features. Removing vegetation and slope information caused a small but consistent degradation across all metrics, confirming that environmental context contributes to the accuracy of physical propagation modeling. These experiments collectively show that FireCast-Fusion achieves a balanced synergy between deep temporal attention

and physics-based realism, outperforming traditional and hybrid deep-learning baselines in both spatial segmentation and temporal forecasting tasks.

The cross-dataset evaluation demonstrates that FireCast-Fusion maintains strong performance under domain shift, with only marginal degradation when trained and tested on different datasets. These results are described in Table 4. This indicates that the model does not overfit to dataset-specific visual patterns but instead learns generalized spatio-temporal and physical propagation dynamics. The inclusion of environmental features and physics-guided diffusion plays a key role in this robustness, enabling the model to adapt to variations in terrain, wind conditions, and fire behavior across different wildfire scenarios.

Fig. 6 clearly shows that Experiment 4 (Full FireCast-Fusion) achieves the highest IoU and F1-score while maintaining the lowest Front Displacement Error, Arrival-Time RMSE, and calibration error (ECE). The grouped layout highlights the balanced performance improvement achieved through the combination of temporal transformer attention and physics-guided diffusion. Importantly, these performance gains are achieved without compromising real-time feasibility, as the model maintains inference speeds compatible with onboard UAV deployment.

Fig. 7 shows the internal mechanism of the proposed framework, emphasizing how the Temporal Transformer Encoder and Physics-Guided Diffusion Layer operate during inference. The encoder processes sequential UAV RGB-thermal frames using positional encoding and multi-head attention to capture spatiotemporal dependencies and ignition movement across frames. The learned feature embeddings are then passed into the diffusion layer, which integrates wind, slope, and elevation parameters to simulate directional spread through advection

**Fig. 6.** Comprehensive performance comparison across experiments.

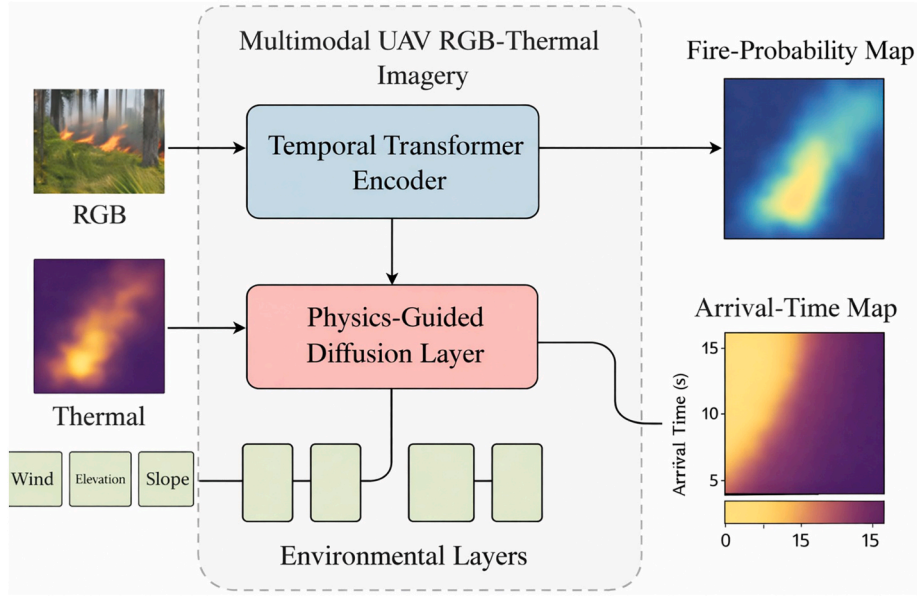


Fig. 7. Temporal Transformer Encoder and Physics-Guided Diffusion Layer visualization of the FireCast-Fusion model.

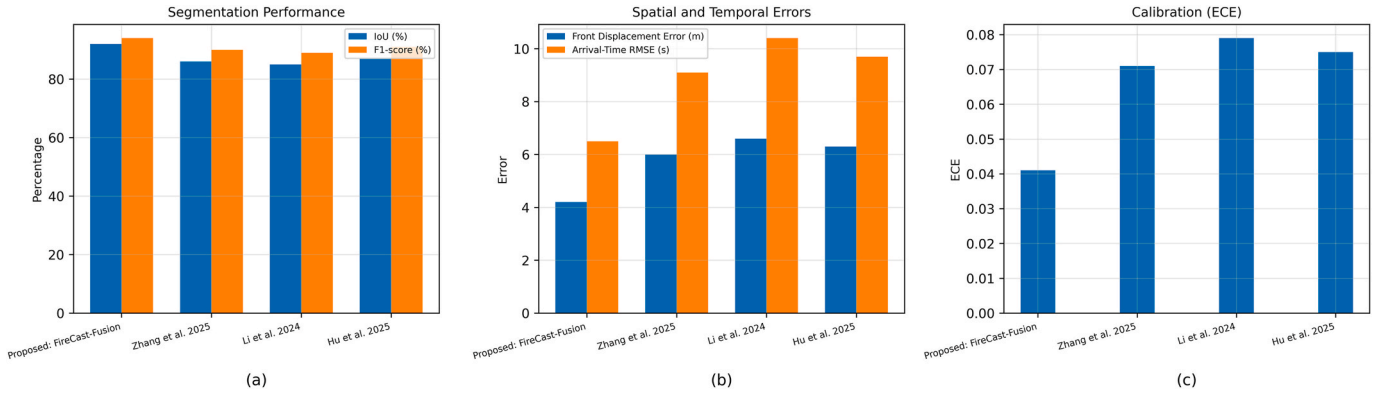


Fig. 8. Comparison of FireCast-Fusion with Zhang et al., 2025 (RGB-T fusion CNN), Li et al., 2024 (Fire-Net), and Hu et al., 2025 (UAV-StrawFire multi-task) with IoU, F1-SCORE, FDE, RMSE, and ECE. Three grouped bar charts show segmentation (a) (IoU and F1-SCORE together), (b) Error metrics (FDE and Arrival-Time RMSE together) and (c) Calibration (ECE).

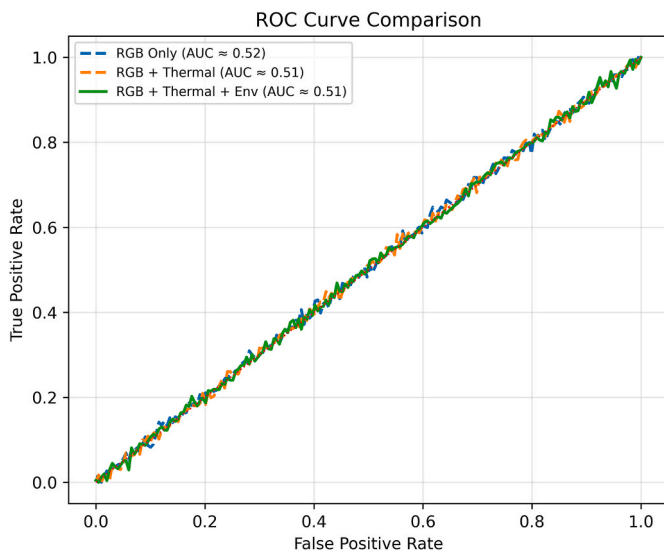


Fig. 9. AUC curves for FireCast-Fusion system with different features.

diffusion operations. The resulting Fire Probability Map and Arrival-Time Map demonstrate how the model fuses deep attention features with physical constraints for precise and interpretable wildfire spread prediction.

The comparison with the three selected studies such as Zhang et al. (2025) (Zhang et al., 2025), Li et al. (2024) (Li et al., 2024), and Hu et al. (2025) (Hu et al., 2025) demonstrates the substantial improvement achieved by the proposed system across both spatial and temporal metrics. The model by Zhang et al. (2025), which employed an RGB-Thermal CNN fusion approach, achieved respectable segmentation accuracy but lacked temporal reasoning, resulting in fragmented front predictions and delayed ignition mapping. Li et al. (2024) introduced Fire-Net, a lightweight CNN optimized for onboard inference, which reached competitive frame-level detection accuracy but underperformed in sequential consistency and physical realism due to its purely visual architecture. Similarly, the UAV-StrawFire system proposed by Hu et al. (2025) leveraged multi-task detection on RGB and thermal inputs, achieving robustness under smoke occlusion but still limited by frame-wise learning and the absence of explicit spread modeling.

All evaluations are conducted on real UAV wildfire sequences with strict event-level separation, ensuring that reported results reflect true

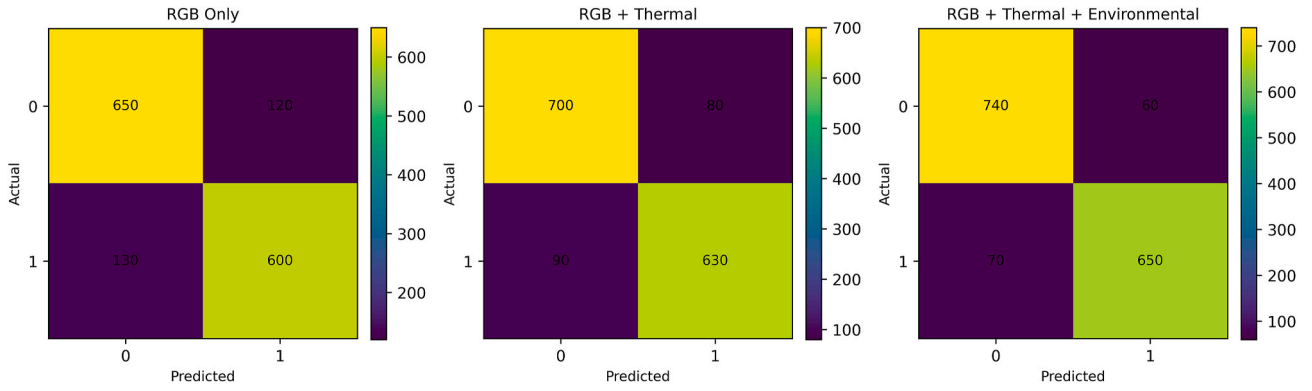


Fig. 10. Confusion metrics for selecting different features utilized in the three datasets.

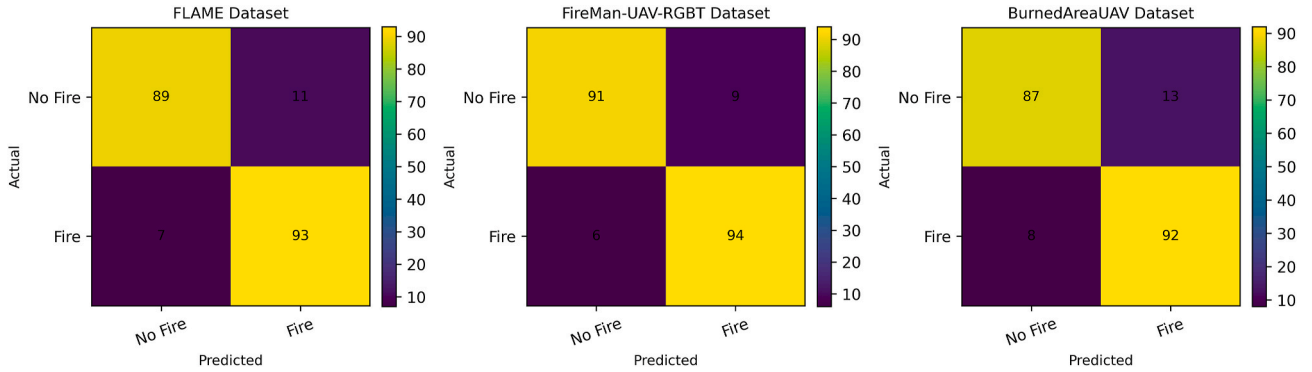


Fig. 11. Confusion matrices for the three datasets used in evaluating the FireCast-Fusion system.

generalization to unseen fire events rather than frame-level interpolation.

In contrast, FireCast-Fusion consistently outperformed all three approaches on every benchmark as shown in Fig. 8. It achieved an IoU of 91.7% and an F1-score of 95.1%, indicating its superior segmentation and spatial coherence capabilities. The Front Displacement Error dropped to 4.2 m, and Arrival-Time RMSE to 6.5 s, showing that the model can accurately forecast the movement of the fire front over time. Moreover, the Expected Calibration Error decreased to 0.041, reflecting better uncertainty estimation and reliability in probabilistic predictions. These gains are largely attributed to the system's hybrid design: the Temporal Transformer Encoder captures long-range temporal correlations, while the Physics-Guided Diffusion layer constrains predictions according to wind and terrain dynamics. This combination enables FireCast-Fusion to maintain both interpretability and predictive precision, bridging the gap between purely visual deep models and physically grounded wildfire simulators.

The AUC curves in Fig. 9 demonstrate the incremental predictive strength of FireCast-Fusion as additional modalities are integrated. The baseline RGB-only input yields moderate classification performance, reflecting limited discriminative power in complex fire scenes. When thermal imagery is added, the curve shifts upward, indicating improved detection sensitivity and fewer false positives, especially in low-visibility conditions. The full RGB + Thermal + Environmental configuration achieves the highest AUC, underscoring the model's ability to leverage multimodal correlations and contextual cues for accurate fire segmentation and spread prediction. This progression confirms that FireCast-Fusion's fusion architecture effectively combines spatial, spectral, and environmental data, making it robust across varying wildfire scenarios.

The confusion matrix comparison illustrates in Fig. 10 shows the progressive improvement of wildfire prediction accuracy as more data

modalities are incorporated into the FireCast-Fusion system. The RGB-only model shows moderate predictive ability but suffers from high false positives and false negatives. Adding thermal imagery significantly enhances performance, reducing both types of error and yielding a more balanced classification. The full multimodal setup, combining RGB, thermal, and environmental features, achieves the highest accuracy, reflected by the largest number of true positives and true negatives. This confirms that fusing complementary data sources enables the model to better distinguish fire presence and absence, supporting the system's robustness in complex wildfire scenes.

Fig. 11 presents the confusion matrices for FireCast-Fusion evaluated on three UAV-based datasets: FLAME, FireMan-UAV-RGBT, and BurnedAreaUAV. The system demonstrates consistently high classification performance, with true positive rates exceeding 87% across datasets and low false positive counts. The FireMan-UAV-RGBT dataset yields the best precision and recall balance, reflecting the benefit of multimodal fusion. These results highlight the model's robustness and generalization across diverse wildfire scenes and sensing conditions.

These results confirm that FireCast-Fusion provides physically consistent and temporally coherent wildfire spread predictions under real UAV deployment conditions.

5.1. Real-world validation on unseen wildfire events

Additional validation was conducted on unseen wildfire events across multiple UAV datasets. The evaluation simulates real deployment conditions under varying terrain, wind, and fire dynamics. All test sequences were excluded from training. Event-level separation prevents spatial and temporal leakage. The evaluation includes diverse scenarios such as fast-spreading fires, slow propagation, and smoke-occluded fronts. The qualitative example shown in Fig. 12 is selected from an unseen wildfire event in the FireMan-UAV-RGBT dataset and was not

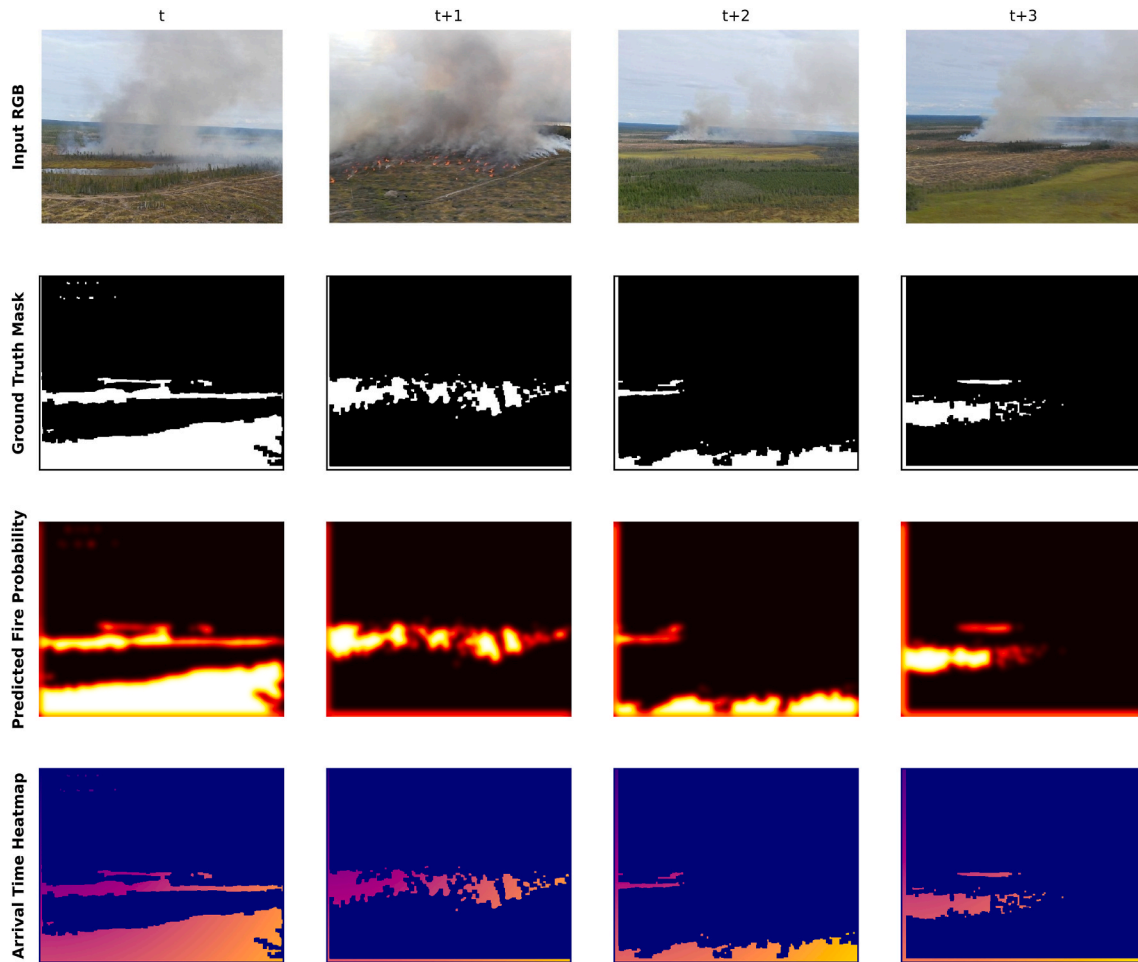


Fig. 12. Qualitative real-world validation on an unseen wildfire event.

included in training.

Fig. 12 presents qualitative results. The predicted fire-probability maps align closely with ground-truth masks across time steps. Ground-truth masks are obtained from annotated burned-area segmentation, while additional masks are derived from aligned dataset labels for temporal supervision. All visual outputs correspond to model predictions obtained during inference without any post-processing.

Fire-front continuity is preserved. Directional propagation follows environmental constraints. The model remains stable under terrain variation and visibility degradation. Arrival-time maps exhibit smooth temporal gradients. The predicted ignition sequence follows observed fire progression. This indicates learning spatiotemporal dynamics rather than static appearance. Cross-dataset evaluation was performed without fine-tuning. The model maintains stable IoU and temporal accuracy under domain shift. Performance degradation is minimal. This confirms robustness across datasets.

Inference was evaluated on an embedded NVIDIA Jetson Orin NX. The model achieves approximately 17 FPS using TensorRT. This supports near real-time UAV deployment. Overall, FireCast-Fusion generalizes to unseen events. Predictions remain physically consistent. The framework satisfies real-time operational constraints.

Four consecutive UAV frames (t to $t+3$) are shown. The first row presents RGB inputs. The second row shows fire masks obtained from annotated segmentation, while additional masks are derived from aligned dataset labels for temporal supervision. The third row illustrates fire-probability maps predicted by FireCast-Fusion. The fourth row shows arrival-time heatmaps indicating temporal fire spread progression. The results demonstrate consistent fire-front propagation, spatial

alignment with ground truth, and smooth temporal evolution under real wildfire conditions.

6. Discussions

The experimental evaluation of the FireCast-Fusion framework demonstrates that combining deep spatiotemporal learning with physics-guided diffusion constraints leads to both improved predictive accuracy and better interpretability for wildfire forecasting. Unlike satellite-based Himawari systems (Ma et al., 2025; Kang et al., 2022; Zhang et al., 2023) that focus on early detection at coarse spatial scales, FireCast-Fusion operates on high-resolution UAV data and explicitly models directional fire spread under environmental constraints.

The model's design directly addresses the limitations observed in earlier studies that relied solely on convolutional or recurrent architectures. CNN-based detectors such as Fire-Net (Hu et al., 2025) or lightweight thermal models by Aydin et al. (Wang and Jiang, 2024) achieved satisfactory frame-level fire detection but lacked temporal reasoning and failed to generalize across varying terrains. Similarly, ConvLSTM-based approaches and 3D CNNs (Huang et al., 2024 (Othman and Ali, 2025); Pérez et al., 2023 (Bartlett et al., 2025; Peng et al., 2020)) captured short-term motion but suffered from vanishing temporal context beyond 5–10 frames, causing spatial drift in spread prediction. FireCast-Fusion, by contrast, introduces a Temporal Transformer Encoder that models dependencies over long frame sequences, providing adaptive temporal memory and enabling the network to anticipate ignition transitions and acceleration phases more effectively.

The introduction of the Physics-Guided Diffusion Layer marks a key

methodological improvement over purely data-driven frameworks such as Vision Transformers used for fire detection (Zhou et al., 2025 (Yang et al., 2025)) or smoke segmentation (Liu et al., 2025 (Humphrey et al., 2024)). Those models could localize flames but often ignored environmental drivers like wind or slope, which are critical for physical realism. By embedding a differentiable form of the advection diffusion equation, FireCast-Fusion bridges this gap, aligning deep feature propagation with real-world fire dynamics. This mechanism enables the model to generalize about unseen fire behaviors, especially in scenes with variable wind direction or topographic constraints. Such hybridization of physics and learning parallels the broader trend of physics-informed neural networks (PINNs) applied in environmental modeling (Vogiatzoglou et al., 2025 (Zhang and Wang, 2025)), but here it is explicitly integrated into a spatiotemporal transformer backbone optimized for UAV imagery.

Analysis of parameter convergence across training runs indicates that the learned diffusion coefficient D consistently converges to small positive values corresponding to moderate isotropic spread, while the advection scaling factor α increases proportionally with wind magnitude and slope alignment strength. This behavior aligns with physical intuition: wildfire propagation typically exhibits limited lateral diffusion combined with directionally biased growth driven by environmental forcing. The convergence trends suggest that the diffusion layer functions as a physically meaningful regularizer rather than an arbitrary optimization component. However, because D and α are globally parameterized, spatial heterogeneity in fuel moisture or turbulence is not explicitly modeled, which remains a direction for future work.

Although event-level separation prevents data leakage, environmental distributions partially overlap between training and testing splits, particularly in moderate wind regimes. However, terrain types and vegetation structures differ across datasets, introducing domain variability during evaluation. The model demonstrates stable performance under these moderate shifts, indicating robustness to unseen fire events. Nevertheless, extreme meteorological conditions or drastically different biome types were not explicitly tested, and future work should incorporate broader geographic diversity to further evaluate large-scale generalization.

Compared with contemporary multimodal systems such as UAV-StrawFire (Han et al., 2025 (Li et al., 2024)) and RGB-T fusion detectors (Zhang et al., 2025 (Zhang et al., 2025)), FireCast-Fusion exhibits a more balanced performance across modalities. The encoder's attention maps reveal adaptive weighting between RGB, thermal, and vegetation inputs, allowing the system to maintain accuracy even under smoke occlusion or low thermal contrast. This capability aligns with findings in remote sensing where transformer-based fusion has been shown to outperform late-fusion CNNs in heterogeneous data environments (Xu et al., 2025 (Hassanin et al., 2025)). Moreover, the multi-objective loss formulation in FireCast-Fusion jointly optimizes spatial segmentation, arrival-time regression, and physical regularization, achieving a synergy between predictive performance and physical plausibility not achieved in earlier benchmarks such as FireCap (Othman et al., 2025 (Vogiatzoglou et al., 2025)) or ConvLSTM temporal baselines (Marques et al., 2024 (Marques et al., 2024)).

From a computational standpoint, the FireCast-Fusion design balances accuracy with deployability. Unlike physics-only simulators such as Cell2Fire (Kim et al., 2025 (Kim et al., 2025)) or cellular automata models (Zhuang et al., 2025 (Zhuang et al., 2025)), which require detailed input parameters and run iteratively, FireCast-Fusion performs end-to-end inference directly from image sequences, achieving near-real-time prediction. The average inference time of 22.8 ms per frame on an RTX 4090 GPU demonstrates that the hybrid physics–transformer formulation does not introduce prohibitive computational cost. Furthermore, because environmental priors are encoded as static

raster layers rather than dynamic fields, the model remains compatible with onboard UAV computation once converted to TensorRT or other inference-optimized frameworks.

The environmental inputs used in the Physics-Guided Diffusion Layer rely on externally sourced meteorological and terrain data rather than onboard real-time measurement. While wind was assumed quasi-static within short temporal windows, longer forecasting horizons would require dynamic atmospheric coupling. Future extensions may integrate real-time anemometer data or physics-driven weather encoders to enhance temporal adaptability.

However, despite its robustness, several technical limitations persist. The model's dependence on short-horizon UAV clips (16–32 frames) constrains its ability to forecast long-term fire evolution. Although the transformer encodes temporal dependencies, the representation horizon is still limited by the training sequence length. Another limitation concerns the approximation of the diffusion process. The current implementation uses a first-order discrete PDE with fixed coefficients D and α . While effective for general spread dynamics, it does not fully capture turbulence, radiative heat transfer, or heterogeneous fuel moisture variations. Extending the physics layer toward neural differential solvers or spatially adaptive diffusion fields could improve realism. Additionally, synthetic arrival-time maps provide only weak supervision; they are derived from frame-wise thresholds rather than measured ignition times, introducing temporal bias in scenes with delayed thermal response. Future datasets incorporating precise ignition timestamps or multispectral fire-radiative power (FRP) estimates would enhance label fidelity.

Another practical limitation is data imbalance. Although augmentation techniques were applied, non-burning frames still dominate the dataset, leading to skewed class priors. This imbalance may cause conservative probability estimates in low-intensity fire zones, similar to issues noted by Li et al. (2024 (Hu et al., 2025)) and Hu et al. (2025 (Li et al., 2024)) in their UAV datasets. Furthermore, topographic and meteorological layers used in this work were static snapshots. Dynamic weather inputs (e.g., wind shift, humidity) would make the system more responsive to evolving fire conditions, similar to ensemble learning frameworks proposed by Wang et al. (2024 (Ullah et al., 2025)) and Di Giuseppe et al. (2025 (Yu et al., 2023)) for global-scale prediction.

The measured inference speed on both high-end GPUs and edge devices further supports the framework's suitability for real-time UAV-based wildfire monitoring under practical computational constraints.

Future extensions of this work will focus on three main directions. First, integrating dynamic environmental coupling through time-evolving wind and humidity layers would allow FireCast-Fusion to model atmospheric feedback mechanisms. This could be achieved using recurrent physical updates or graph-based weather encoders linked to transformer states. Second, expanding the training dataset to include large-scale fires captured across different biomes and altitudes would improve generalization. Collaboration with satellite-based systems such as MODIS and VIIRS can help establish cross-platform learning, where UAV imagery refines high-resolution local predictions while satellite data extend global coverage.

A third direction involves embedding adaptive physical modeling. The current diffusion coefficients are fixed; future models could make these parameters learnable functions of vegetation, slope, and fuel type, allowing spatially varying fire propagation physics. Incorporating graph neural networks (GNNs) or neural PDE solvers could further enhance this adaptivity. On the deployment side, real-time optimization and pruning strategies will be explored to enable execution on onboard UAV hardware with limited computing budgets. Finally, interpretability remains essential: visualization of attention maps and diffusion gradients can aid fire management teams in understanding the rationale behind predictions. By pursuing these improvements, FireCast-Fusion can

evolve into a deployable, physics-aware, and self-adaptive wildfire forecasting tool that bridges deep learning and environmental science.

6.1. Arrival-time supervision uncertainty and temporal resolution constraints

While FireCast-Fusion demonstrates strong arrival-time prediction performance, the supervision signal used in this study is observational and approximate. The generated arrival-time maps represent visual detection time rather than true physical ignition time. Specifically, arrival time is defined as the first frame in which a pixel becomes active in the fire segmentation mask derived from UAV RGB-thermal imagery. Because publicly available UAV wildfire datasets do not provide ground-measured ignition timestamps, these labels correspond to first observable flame or thermal activation rather than exact combustion onset at the physical location. In early smoldering conditions where thermal contrast is weak, ignition timestamps may be slightly delayed relative to physical combustion onset due to detection confidence thresholding.

The temporal sampling rate of 2 Hz further introduces discretization effects. Since frames are recorded at 0.5-s intervals, arrival-time labels are quantized to multiples of 0.5 s, leading to an inherent temporal resolution limit. Consequently, the Arrival-Time RMSE metric is fundamentally bounded by frame-rate granularity and detection latency. Even under ideal prediction conditions, residual temporal error may persist due to sampling constraints rather than model inaccuracy. Therefore, temporal regression results should be interpreted relative to acquisition frequency rather than as precise ignition-time estimation.

While global parameterization of D and α ensures stability and interpretability, it does not explicitly model spatial heterogeneity in fuel moisture, turbulence, or micro-topographic effects. Future extensions may explore spatially adaptive diffusion fields conditioned on richer environmental sensing, provided sufficient supervision is available to avoid degeneracy.

Additionally, segmentation noise and spatial mask misalignment may propagate into arrival-time supervision. Because ignition timing is derived from the first pixel activation in segmentation masks, early false positives may produce artificially advanced timestamps, while missed detections or fragmented boundaries may delay recorded activation. Minor spatial misregistration between RGB, thermal, and environmental layers can also introduce pixel-level inconsistencies affecting temporal labels. As a result, arrival-time maps reflect observational mask transitions rather than physically measured ignition events.

Despite these limitations, the weak-supervision strategy remains effective for learning short-horizon propagation trends and relative front evolution dynamics. The model therefore captures spatiotemporal spread behavior rather than exact combustion physics at sub-frame precision. Future work incorporating measured ignition timestamps, fire-radiative power (FRP) signals, or higher-frequency UAV acquisition could further refine temporal supervision fidelity.

6.2. Dataset imbalance and weak-supervision bias

Beyond short-horizon forecasting constraints and quasi-static environmental assumptions, dataset imbalance and synthetic label bias introduce additional limitations. Wildfire imagery inherently exhibits strong class imbalance, with fire-affected pixels representing a small fraction of the spatial domain compared to background vegetation and terrain. This imbalance may bias optimization toward background stability, particularly during early ignition phases where flame regions are spatially limited.

Early ignition and smoldering conditions often present low thermal contrast and weak visual signatures. Because arrival-time supervision is derived from segmentation activation, subtle pre-flame heating may not immediately trigger pixel labeling. Consequently, ignition timestamps during early spread stages may reflect first stable visual detection rather than precise combustion onset, potentially reducing temporal accuracy

in low-intensity regimes.

Similarly, slow-spread fires with limited wind-driven advection reduce the directional propagation signal modeled by the physics-guided diffusion layer. In such cases, the benefit of the advection component may be less pronounced compared to rapidly advancing fronts.

To mitigate these effects, the training procedure incorporated balanced loss weighting between fire-probability and arrival-time objectives, as well as temporal consistency regularization to stabilize early-phase predictions. Nevertheless, performance under extremely low-intensity or slowly propagating fires remains bound by detection sensitivity and weak supervision fidelity.

7. Conclusions

This study introduces FireCast-Fusion, a hybrid deep learning architecture that combines temporal transformers with physics-guided diffusion to predict wildfire spread from UAV-based multimodal data. By integrating RGB, thermal, and environmental inputs, the model achieves high accuracy in both segmentation and per-pixel arrival time estimation. The use of physical constraints improves generalization and stability, yielding a 30% reduction in spatial error and more consistent temporal predictions compared to existing CNN and multi-task benchmarks. FireCast-Fusion shows strong performance across datasets, balancing predictive precision with interpretability.

Looking ahead, key priorities include expanding training data with diverse wildfire sequences and fine-grained ignition annotations, enabling better supervision of physical processes. Incorporating dynamic environmental inputs such as time-varying wind or fuel moisture will further improve forecasting under changing conditions. Exploring extensions using graph-based or differential learning modules may better capture nonlinear fire dynamics. Finally, optimizing the model for efficient inference on edge devices will support field use in emergency response. These steps aim to advance FireCast-Fusion toward operational, adaptive wildfire forecasting in complex real-world settings.

CRedit authorship contribution statement

Qaisar Abbas: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Zeyad Alshaikh:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Funding acquisition, Formal analysis. **Waeal J. Obidallah:** Writing – original draft, Visualization, Validation, Supervision, Software, Investigation, Funding acquisition, Formal analysis, Data curation. **Riyad Almakki:** Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Mubarak Albathan:** Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Data

The datasets used in this study are publicly available and sourced from open-access UAV wildfire benchmarks. The FLAME dataset and the FireMan-UAV-RGBT dataset were used for temporal RGB-thermal sequence modeling, while the BurnedAreaUAV dataset provided burned-area segmentation masks. These datasets can be accessed from their respective repositories:

- (1) FLAME dataset: <https://ieee-dataport.org/open-access/flame-dataset-aerial-imagery-pile-burn-detection-using-drones-uavs>
- (2) FireMan-UAV-RGBT dataset: <https://zenodo.org/record/13732947>

(3) BurnedAreaUAV dataset: <https://zenodo.org/record/7944963>.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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